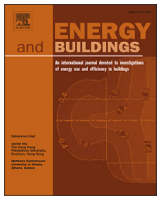
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Identification of representative building archetypes: A novel approach using multi-parameter cluster analysis applied to the Swedish residential building stock

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ABSTRACT

Building archetype identification is crucial for Urban Building Energy Modeling (UBEM), but is still considered one of the biggest challenges in this field. New methods of data acquisition, along with data mining techniques such as clustering, have recently received attention for the possibility of significantly increasing identification reliability and archetype accuracy. This paper aims to establish a new and simple clustering methodology for developing building archetypes for hybrid UBEM, using open data sets and multiple diverse variables, that is still reliable and possible to validate without the use of metered energy use or real building data. The methodology uses *k*-means clustering for 10 building parameters simultaneously, including socio-economic parameters obtained using spatial interpolation of statistical values. Building archetypes are successfully developed for the residential building stocks of two case study areas in Sweden. The results also show that the error metric values for multiple iterations diverge after a certain number of clusters, even when using the same clustering methodology on the same data set. This discovered effect, along with the combined use of one well-known and one novel error metric, constitutes a framework well adapted to accurately determining the optimal number of building archetypes.

1. Introduction and background

The building sector plays an integral role in the energy systems of our present society, where the energy use in households currently accounts for 28% of the total energy use in the EU [1]. Urban areas account for more than 70% of global carbon dioxide emissions associated with energy use [2] and with the current overall population growth, urban population will have increased by 50% by 2050 as urbanisation increases worldwide [3]. The EU has set ambitious targets for mitigating climate change where the built environment in cities is expected to play an essential role, including reducing greenhouse gas emissions by 40% and improving energy efficiency by at least 32.5% compared to 1990's levels [4]. In response to the growing urbanisation rate, and with the prospect of climate change, it is necessary to develop urban planning towards more sustainable strategies and to further include energy planning with interconnected solutions [5]. The need for implementing integrated energy planning in cities along with the key role of buildings in the urban energy balance has led to the emergence of city-scale building-oriented studies [6].

1.1. Urban Building Energy Modeling and archetype development

Detailed simulations of the energy demand in buildings are essential for well-informed policy decisions regarding sustainability measures or strategies for energy transition. With well designed models, accurate predictions can be made. Scenarios regarding proposed measures or the impact of external factors on the energy system can be simulated with high precision, which could be of high importance when determining future building energy performance or designing new urban energy systems [7].

In Building energy modelling (BEM), the fundamental approach is to quantify the energy performance of the building for different spatio-temporal resolutions. The energy performance is determined by multiple factors, such as building characteristics, occupant behaviour, climate and location. For city-scale applications, simulating thousands of buildings at a high spatio-temporal scale quickly becomes data-intensive and computationally heavy. This is a challenge for city-scale building energy simulations, that is commonly addressed by applying several different grouping and clustering techniques. Instead of data collection and model characterisation for each individual building, a

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subset of representative buildings, i.e. building archetypes, can be created which are later used for upscaling. This abstraction of the building stock is a useful and often necessary approach for large-scale applications, and has also been proven useful regarding other functions such as model transferability, compression or explanation of information, and energy performance benchmarking [8].

Using archetypes can however increase the risk of model oversimplification, and is currently associated with the largest uncertainty for building energy simulations [9,10]. Further, an assumption made when using archetypes is that they are representative for all building parameters, including for instance construction materials, U-values and electricity use, and not only the parameters present in the segmentation or clustering process.

City-scale, bottom-up engineering based modelling of heat and mass flows to simulate the energy performance of buildings is commonly referred to as Urban Building Energy Modeling (UBEM) [9]. In recent UBEM research the hybrid approach is however becoming increasingly common, rather than the purely bottom-up (physical) approach. In hybrid models the buildings are modelled after their physical characteristics, while data such as building air leakage or occupancy, which are typically stochastic parameters, are being treated as in statistical models. This creates empirical distributions of the uncertain parameters and therefore the model can make use of both deterministic and probabilistic processes. Hybrid models solve many of the shortcomings of physical models, but previous literature reviews argue that there are still some significant knowledge gaps that should be subject to further research, particularly archetype development [6,7].

The process of archetype development typically includes three steps: classification (or segmentation), characterisation, and calibration.

Classification involves categorising the buildings into sub-groups to better represent the differences in typology or simulation behaviour. Typical parameters for segmentation are building age, use, size, or similar. After classification of the building stock, the identified archetypes need to be characterised by all relevant building parameters, not only those used for segmentation. These parameters are usually non-geometric, which might include construction materials, occupancy, and heating, ventilation and air conditioning (HVAC) systems.

For classification and characterisation of archetypes there are also three major approaches: deterministic, probabilistic or using cluster analysis. With a deterministic approach, a building is classified according to its theoretical energy use which is directly determined by input parameter values. This categorisation is simple to perform with general building data and knowledge of building physics. However, this approach can be too simplified and generic and misrepresent the real diversity of the building stock, which may result in inaccurate energy demand patterns [6]. Building archetype characteristics are typically represented by a set of fixed values per archetype or with values linearly dependent on geometric parameters.

A probabilistic approach utilises real data, i.e., metered energy use or building statistics, to identify the building parameters with the strongest correlation to the energy use intensity. Building archetype characteristics can be represented by probabilistic distributions instead of fixed values. Statistical approaches are easy to implement and can accurately predict the energy demand for a large building stock. However, they have a high dependence on data availability and are poor at noticing patterns in the building stock, as statistical models do not provide representation and disaggregation down to individual building level [10].

Cluster analysis techniques instead start from noticed patterns, providing an unsupervised classification of the data according to their similarities. Multivariate cluster analysis (MCA), or just clustering, is a central and useful data-mining technique concerned with the identification of homogeneous groups within a data set [11]. With MCA techniques, many building features can be analysed in parallel and with a more exact segmentation of the building stock compared to traditional classification procedures [12]. They do not impose particular weights on some variables, producing an un-biased grouping result. Using cluster analysis for building archetype development is rapidly growing in UBEM and is a promising area for further improvements in classification and model accuracy [7].

1.2. Cluster analysis

A multitude of clustering methods and algorithms have been presented, with four categories being most common in the literature: partitional clustering, hierarchical clustering, density-based clustering, and grid-based clustering [13]. Two of the most commonly used algorithms are the *k*-means algorithm, which is a partitional clustering method, and the Agglomerative hierarchical clustering (AHC) method.

The *k*-means algorithm is complex but computationally fast. It is based on Euclidean distance, which is usually successful in finding clusters even with a high number of clustering parameters; however, it is not capable of analysing arbitrarily shaped clusters [13]. Ghiassi and Mahdavi [11] analysed a multitude of scenarios and successfully created accurate representative buildings with up to 12 clustering parameters. For identification of representative buildings, Tardioli et al. [14] evaluated multiple clustering algorithms with different normalisation approaches and validation indices. They found that *k*-means was the method that performed best for residential multi-family buildings. The algorithm has also been proved to perform better than AHC to identify sub-groups of energy consumption in buildings [15] and for thermal zoning and load pattern analysis for office buildings [16]. Apart from finding representative buildings, *k*-means has been demonstrated to be a useful algorithm for segmenting energy use groups [17]. Evaluating clustering-based methods for simulating the annual heat demand or peak heat demand, De Jaeger et al. [18] found that *k*-means clustering combined with re-scaling standardisation techniques yielded the lowest root mean square error for all key performance indicators tested, indicating that it gives the best clustering result. Gangolells et al. [19] successfully used the algorithm to find representative buildings using an Energy performance certificate (EPC) database for a large set of office and mixed use buildings, gaining highly representative archetypes. Hsu [15] analysed and compared recent work using clustering techniques to identify sub-groups of energy consumption in buildings. The study concludes that *k*-means gives less accurate predictions than cluster-wise regression but more stable clusters when the value of k is chosen, and that there is a trade-off between prediction accuracy and cluster stability. The value of k is however chosen arbitrarily, without any underlying reasoning or methodology, which may undermine this conclusion. One of the most common problems faced when using *k*-means clustering is to determine the optimal number of clusters, as the clustering algorithm has the desired number of clusters as an input value, not as a result [11,13,17,18].

1.3. Data for UBEM and clustering approaches

The data that can be used for bottom-up, hybrid UBEM mainly requires four types of data inputs: geometric data, non-geometric data (including usage patterns and statistical data), weather data, and validation/calibration data [20].

The access to real/metered energy data for validation and calibration is often restricted [20,21] resulting in most UBEM studies lacking proper model validation [6]. In a comprehensive taxonomic review of UBEM workflows, Malhotra et al. [22] found that only 44% of studies validate the outputs using measured data and 30% does not perform any validation at all. Access to metered energy use values or other restricted data makes the model better tailored for a specific location. This affects the modelling process's ability to be reproducible or transferable to other locations. This is a common issue in UBEM, where the overwhelming majority of reviewed publications cannot be reproduced as data sources and simulation models are not openly available, and that the process is not described in detail [22]. Limiting the data sources

to public data sets and records makes the model less specific and less sensitive to characteristics of given data. One major advantage is then the ability to apply the model for any location, since the data should be equally available and accurate for the whole country (if using public national data sets).

For buildings within the EU, there are openly available databases for archetype or reference buildings and energy-related indicators, such as TABULA [23] and ODYSSEE [24]. However, they are deterministically defined and contain data on a national level, which disregards important information regarding local building stocks. Reliance on deterministic assumptions simplifies parameters that are particularly relevant in energy modelling and assumptions considered for national level building stocks may not hold true for district level archetypes, which can lead to results that misrepresent urban energy demands [25].

In a review of common UBEM applications, evaluating research gaps and examining auxiliary model inputs, Dahlström et al. [7] found several areas for further potential developments in UBEM, especially for building archetype classification. Among them are advances in data gathering from comprehensive databases such as the EPC database, and increasing the number of classification parameters, especially through clustering processes. It was also found that including socio-economic parameters, such as household income and demographics, is commonly mentioned as a possible improvement for UBEM, but is mostly absent in the literature.

Energy Performance Certificates (EPCs) are public domain documents allowing energy use comparison in the building stock and identification of possible energy efficiency improvements [26]. In most EU countries, energy performance assessments are based on calculations derived from building parameters and statistical data, along with some measured values. A notable exception is Sweden, where most collected EPCs are assessed using measured values [27,28]. The main focus of the Swedish EPCs is to provide data on building energy use for heating, cooling and tap water, and to provide proposals for possible energy efficiency measures [29]. The energy use is presented as measured values per energy carrier, total energy performance (as kWh/m^2), as well as building energy class (based on primary energy conversion factors). Additionally, the EPCs contain data on building use type, construction year, total heated area, address and location, ventilation system installed, electricity production (i.e. solar panels) and corresponding metadata.

The interest for EPCs is currently growing in governing bodies, most notably in the EU. A recent proposal to the European Commission calls for the EPCs to be updated by 2025 [30], and to ensure comparability across the EU, all EPCs shall be based on a harmonised scale of (new) energy performance classes and comply with a common template. Additionally, the proposal aims to increase the EPC coverage across the Union. Each member state shall set up a national database, and the obligation to have an EPC is extended to all public buildings, buildings with renewed rental contracts and buildings undergoing major renovation [30].

Consequently, EPC databases are one of the most extensive data sources for building energy performance currently available, especially in the countries of the EU and in the UK [31]. EPCs are often issued by governmental agencies and are mandatory if certain criteria are met. As a result, EPCs are rapidly growing as a tool for decision-makers and energy modelling researchers among others [8,32,33].

There has been some criticism regarding data quality and content [8] as well as of the accuracy of the methods for deriving the heated floor area [28] but the database is continuously updated, and the uncertainties have recently been reviewed and assessed by von Platten et al. [31]. In Sweden, EPCs are mandatory since 2006 for newly constructed buildings, if the building is larger than 250 m^2 or if the building is made available for sale [26]. The EPC coverage, measured in EPCs per capita, varies from country to country and is currently highest in the UK, Ireland, and Denmark, where Sweden is also in the top category [34]. Although this coverage is increasing, the issue remains that there are still many buildings currently in use that are missing in the EPC data set. For buildings that do have EPCs, archetypes are needed for characterisation of properties that are not in the EPCs but are required for energy simulation of the buildings, such as the construction of building elements, materials used, etc. The basic idea behind the approach of this study is the assumption that buildings that are grouped together by similarity in certain respects, e.g. by the method proposed in this paper, will also be similar in these other respects, so that the missing properties can be identified from the archetype buildings.

1.4. Aim, research gaps and contributions

The use of clustering methods for building archetype development in UBEM is an emerging field of research. A core task is to improve model quality and accuracy while still ensuring low computational cost and low data collection difficulty. Determining the optimal number of clusters is also crucial when using clustering methods, especially *k*-means. Recognised and advanced methods to do this exists, but a simple and cost-effective yet precise method would be preferable. The overall procedure needs to be simple and reliable, which can be achieved by expanding on simple methods and by detailed evaluation for reliable results. There is also a lack of exploration of clustering parameters; many parameters have not been tested in clustering analysis for archetype development applications, including socio-economic parameters. As the problem of acquiring metered energy use data or real building characteristics for archetype development and/or validation is still a barrier for UBEM studies, accurate models that can run on only open data sets are desired.

This paper aims to bridge these knowledge gaps by proposing a new simple clustering methodology for developing building archetypes for hybrid UBEM, using open data sets and multiple diverse variables, specifically including socio-economic parameters. The process should be reliable and possible to validate without the use of metered energy use or real building data. With an easily adaptable and efficient modelling methodology, using only open data, the procedure could be widely applied by decision-makers, researchers or other stakeholders to create reliable and accurate UBEMs with considerable less time and effort. Building footprint geodata and EPC data are available for the countries of the EU and the UK [34,35], and statistics for socio-economic indicators exist but is likely varied in extent and presentation between each country. This should enable the methodology to be utilised for most European building stocks. The novelties and main contributions of this paper are as follows:

First, we extend the use of k-means and the so-called elbow method for identifying the optimal number of clusters. In particular, we present the novel use of cluster “distinctness” and the use of a divergence point for analysing the maximum number of stable clusters. Secondly, we apply clustering simultaneously to building geometric parameters, heating systems and socio-economic factors, which has not been done previously in the available literature. Thirdly, as shown in the review paper by Malhotra et al. [22], 95% of UBEM studies cannot be reproduced, which is a major issue for the field. This work utilises open data sets and various data-matching techniques to get around this problem, paving the way for later UBEMs that are accurate yet fully reproducible. It should be noted that the proposed method is limited to the first important step in archetype-based UBEM, the identification of representative archetype buildings. This is done from data that are openly available for all buildings. Further steps in the modelling process, such as detailed characterisation of the representative archetype building (which requires collecting detailed building information for the chosen representative building) and modelling and simulation of buildings based on the archetypes (which requires a validated UBEM), are outside of the scope of this paper. Therefore, the main result of the paper is the proposed clustering-based method and not the resulting archetypes themselves, nor application of them in UBEM simulations. The emphasis of the paper is on the methodology, while analysis, description and

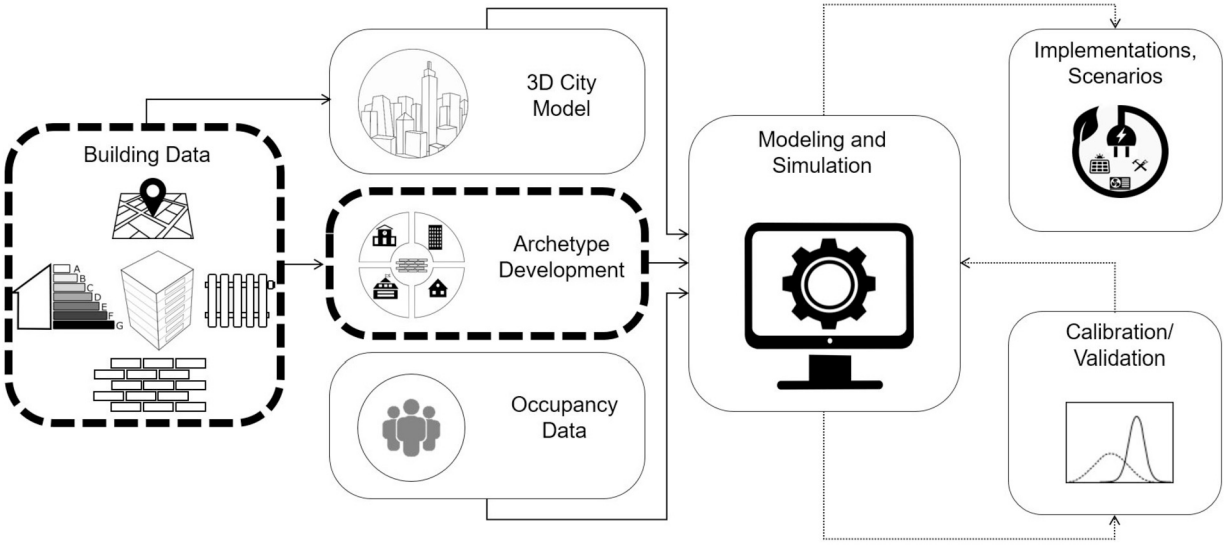


Fig. 1. Overview of the UBEM work flow with the Building Data and Archetype Development sections highlighted.

characterisation of the resulting archetypes are covered more briefly and mainly left to future studies where the archetypes are applied in UBEM.

1.5. Structure of the paper

In this paper, representative building archetypes are identified using a multi-parameter clustering approach. The *k*-means clustering algorithm is applied over a large data set including data retrieved from several open-data sources. Two case study areas are used for the study, with highly different building stocks. The two cases are used to determine the accuracy of the procedure, but also to test the effectiveness of applying the methodology for a new building stock. In this work, all stages of the identification and clustering process are considered. First, the clustering parameters are chosen based upon a literature review. Building data are collected from several open-data sources and are subject to relevant pre-processing, before all buildings are assigned parameter values by combining these data sets and using spatial interpolation of aggregated data. After deciding upon which cluster analysis algorithm to apply, the problem of determining the optimal number of clusters is addressed, which directly influences the resulting number of building archetypes when using a clustering approach. Finally, the accuracy of the methodology is evaluated using the appropriate error metrics, and introducing a novel metric for analysing how representative the identified clusters are.

The remainder of this paper is structured as follows. Section 2 presents the data used in this study, describes the sources and associated pre-processing steps, and subsequently describes the study methodology clustering procedure. The results and analysis of the results and the identified archetypes are presented in Section 3. Section 4 provides a discussion of the findings, followed by a wrap-up and conclusion in Section 5.

2. Methodology and data

This section describes the procedure for cluster analysis used in this paper, including data, case studies and clustering methodology.

Section 2.1 introduces the case studies and data sources used in this study. The parameters later included in the clustering process, and how they were processed, are presented in Section 2.2. How the parameters were included in the clustering algorithm, by assigning them to buildings, is explained in Section 2.3. Section 2.4 describes the subsequent data cleaning procedures. The clustering algorithm and how it works is later presented in Section 2.5. How to determine the optimal number of clusters for the algorithm is described in Section 2.6. The evaluation and validation of the clustering results, using different metrics and indices, is included in Section 2.7. A general UBEM framework and work flow is presented as a flowchart in Fig. 1, with the sections covered by this study highlighted with weighted and dashed outlines. A schematic overview of the research methodology and work flow of this study is shown in Fig. 2.

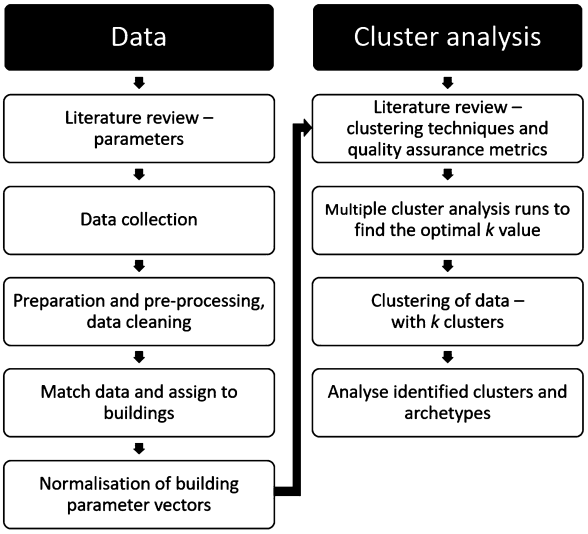


Fig. 2. Overview of the research methodology.

The method is based on using data that are openly available for each building and applying the proposed extended *k*-means clustering analysis to identify clusters of buildings that have similar properties, and within each cluster find one archetype building that is representative of the cluster. The basic premise or assumption is that if the archetype is representative of the cluster in terms of the analysed parameters, then it will also be representative in terms of other characteristics (construction details, U-values, etc.) that are not available for all buildings but still needed in order to fully characterise, model and simulate the buildings within an UBEM. Thus, instead of having to characterise each individual building in detail, this has to be done only for the identified archetype buildings. As Fig. 2 shows, the process analysed in this paper is limited to the clustering and archetype identification, whereas fur-

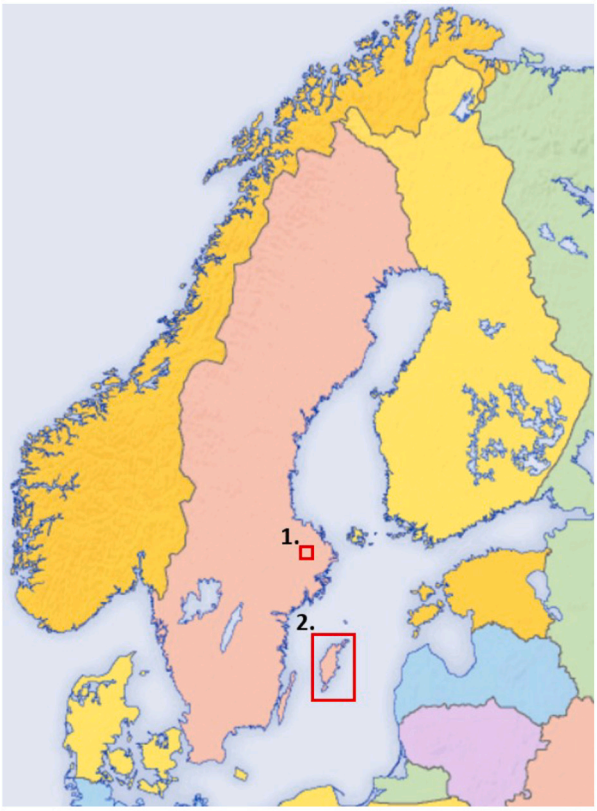


Fig. 3. Geographical location of 1. Uppsala and 2. Gotland within Scandinavia.

ther characterisation, modelling and simulation will be approached in future research.

2.1. Case studies and data sources

For this paper, two case studies were chosen to be able to compare the results of the analysis and to analyse the relative difficulty of using the procedure for a completely new data set. The first area is the city of Uppsala, Sweden, and the second area is the island of Gotland, Sweden's largest island. The island of Gotland is mostly rural, with one city and some smaller towns and villages. Rural areas are not generally included in UBEMs and including areas greater than just urban areas is not discussed in literature. The main difference usually stated is the diversity of the building stock, and the Urban Heat Island effect [6], which creates a different micro-climate in urban areas compared to rural. Being an island, the Gotland area has a defined system boundary, creating the opportunity to include all buildings and gather data for the whole region. For a city like Uppsala, the area to include in the model is inherently more arbitrary; the included area for data acquisition can be defined either by pre-made definitions, like postal areas, or manually by drawing a (subjective) boundary. The included case study areas and their respective geographical location within Scandinavia is shown in Fig. 3.

All buildings with a postal code that matched the case study area in the EPC data set were included in the study. The focus in this paper is on residential buildings due to insufficient sample size in the EPC database (industry buildings are not required by law to have an EPC in Sweden [26]) and higher complexity for non-residential or industry buildings. Residential buildings are often further divided into single family housing (SFH) and multifamily housing (MFH) because of their inherent considerable differences, a convenient categorisation that is also used in this paper. The number of included buildings divided by building type is presented in Table 1, where miscellaneous buildings include mixed-use and non-residential buildings, for example restaurants and schools. The total number of residential buildings is 7042 for the Uppsala data set, and 4098 for Gotland.

Table 1
Number of buildings in the EPC data set.

Category	EPC, Uppsala	EPC, Gotland
SFH	3723	2861
MFH	3319	1237
Industry	141	17
Misc.	923	929

Table 2
Grouping for resident age based on statistical data.

Group	Resident age span
1	0-6 years
2	7-15 years
3	16-19 years
4	20-24 years
5	25-44 years
6	45-64 years
7	65+ years

Cadastral data, or geographic information system (GIS) data, were obtained from the Swedish Mapping, Cadastral and Land Registration Authority [36] for properties and buildings. The GIS dataset is assumed to include all known buildings present at the time for data collection. The relevant cadastral data obtained includes building and property footprints and coordinates, with corresponding property address and building type. The division by typology is done in a similar but not exact same way as for the EPC dataset, which makes comparison by building type between the two data sets difficult. A high percentage of MFH and misc. buildings have EPCs, while the EPC coverage of SFH is low. This is especially true for Gotland, due to the large numbers of SFH in a rural context, including summer houses, small farmsteads etc. which are not required by law to have an EPC [26].

For statistical data, factors such as income, education, home ownership and sometimes also gender, ethnic background or similar can collectively be referred to as socio-economic data or indicators. In Sweden, socio-economic and statistical data are available from the national agency Statistics Sweden (SCB) [37]. Data that can be acquired for a geographic area include resident age, gender, marital status, education, employment, income and total population. The data are aggregated geographically and anonymised in the database, so as to be considered open and non-sensitive. Resident age is assigned in this data set ordered by age group, as described by Table 2. For cadastral and statistical data, no area division based on similar borders was available. For simplicity, all data within a coordinate rectangle that enclose all buildings in the EPC data set was included (See Fig. 8).

2.2. Clustering parameters

2.2.1. Building characteristics

The EPCs contain information about building construction year, heated area (building area heated to $> 10\,^{\circ}C$), energy use (for heating and electricity) and energy performance (kWh/m^2), all of which can be used as parameters for cluster analysis. For building energy use, the EPCs account for the amount of energy used for heating in kWh/yr. Nearly all values are metered or calculated from annual fuel consumption, which is then standard year-corrected to better represent the long-term characteristics of the building. With standard year-correction, the normal climate in the locality of measurements is compared to how the climate has been during the particular period for which there are measured values, and corrected using an Energy Index issued by the national weather authority [38]. The national weather authority in Sweden, SMHI, has recently updated the standard year period by moving it

Table 3
Building heating systems.

Heating system main group	Included heating systems and sources
District heating	District heating District cooling
Boilers	Oil Gas Firewood Wood chips/pellets Misc. biofuels
Electric heating	Water-based electric heating Direct electric heating Forced air electric heating Comfort cooling Building electricity
Heat pumps	Ground-source heat pump Air-source heat pump Water-source heat pump Exhaust air heat pump

forward 10 years. This will cause differences from previous EPC energy values as the mean temperature has increased by 0.5 °C during this period [39]. For energy performance the corrected values for heating are divided by the heated area. The energy use values are reported for each type of heating system, which were divided as shown in Table 3. Using EPC data, the buildings can be categorised after which heating systems they use; however, as many buildings do not exclusively use one source of heating but rather a mix of several, it is of no use to deterministically categorise the buildings after heating system. One solution is instead to calculate the share of each heating system used for each building. Heating of hot water is included but assumed to be heated by the same share of heating systems. An example for a rural SFH could then be 0% district heating, 50% boilers, 10% electric, and 40% heat pump.

Building volume, often simplified as building footprint area multiplied by the height, is often used as a model parameter. In a study using multivariate cluster analysis for UBEM using archetype buildings, Ghiassi and Mahdavi [11] show that including building size as a classification criterion relatively increased predictive performance when comparing otherwise similar scenarios. To estimate the height of buildings, in most existing UBEMs, the 3D representation of the building morphology, referred to as 3D city models, is the most reliable source of information. Yet, in the absence of these comprehensive 3D city models generated, simpler versions of these types of models have to be made from scratch. The common method for making 2.5D or simplified 3D models of buildings is to make use of LiDAR point clouds to approximate the elevation of buildings from the ground and extrude the building footprints accordingly. In this study, using an open-source tool, 3dfier [40], and available data on building footprints, and LiDAR point clouds, 3D shoebox models of building stocks in Sweden were created. The available national LiDAR data in Sweden is considered as low-resolution where the point density is only 0.5-1 point/m^2. Therefore, for some of the buildings, especially those partly or completely over shadowed by another object, such as a tree, the actual height of the building cannot be properly captured from the low resolution point clouds. This issue can be occasionally seen in single-family buildings. Therefore, a post-processing method for altering the estimated height from low-resolution data was used. In this multi-step method, the estimated height is first compared to the average height of similar building types in the surrounding area. If the error is bigger than the chosen threshold of 30%, then the height is recalculated based on the average height as well as the available information on the number of floors, heated floor area and the footprint area of a building. The choice of 30% as a reasonable error threshold was chosen as a trade-off between calculation complexity and accuracy.

2.2.2. Statistical data

Socio-economic factors such as resident income, gender and age (see Section 2.1) have been proven to affect domestic electricity consumption [41,42], energy demand for heating [43,44], per capita electricity consumption [45,46] and building energy performance [47]. Total population can also be converted into population density, which has a negative correlation with CO_2-emissions [2] and energy use [48] per capita. This effect however also relates to resident income; in dense wealthy areas, in for example the USA, population density instead has a positive correlation with CO_2-emissions [2]. Socio-economic factors are directly tied to building occupant presence and behaviour, which in turn largely influence the energy demand of buildings [7,49]. The insufficient use of occupant behaviour models is one of the major causes of the “performance gap” in UBEM, i.e. the difference between simulated and observed energy use [49].

Statistical and socio-economic data are publicly available as aggregated data. Detailed and building-specific data would be more useful and provide more accurate results; however, aggregated data is necessary to use since the GDPR European Union law [50] applies to personal data, i.e., information that is related to an identified or identifiable individual. For acquiring personal data in Sweden, an ethics review process is necessary, which is resource- and time-consuming and the obtained data would not be considered open-source. Also taking into account the difficulty of acquiring metered energy use data, designing an UBEM framework consisting of only publicly available data sets is a more compelling and practical approach.

Resident gender was included as one of the socio-economic building parameters obtained for cluster analysis. However, the data proved to be the most homogeneous of the parameters, with most values being within a close distance to the mean. If the majority of clusters have a mean value for the share of gender that is close to the mean of the whole data set, the division into clusters for this parameter becomes redundant. Therefore, it was decided to not use resident gender as a parameter for the final results of the study.

2.3. Assigning data to buildings

The selected parameters were:

1. Construction year
2. Building volume [m^3]
3. Energy performance [kWh/m^2,yr]
4. Income per person [kSEK]
5. Resident age (group, mean value)
6. Population density [ppl/km^2]
7. Share of district heating
8. Share of boilers
9. Share of electric heating
10. Share of heat pumps

Connecting EPC data to GIS data requires some fine-tuning. The EPCs do not include geographic coordinates and the GIS data does not include building postal addresses. Another issue is that the GIS data set contains all buildings known, up to the date of data retrieval, but the EPCs cover only a percentage of the existing building stock (see Section 2.1), which creates a discrepancy in data coverage. Our solution was to make EPCs the first priority data set, and assign other values based on the EPCs. One way to easily connect the two data sets is by property address, since the property address is present in both data sets. One property can contain multiple buildings but a majority of properties (90% in Gotland and 84% in Uppsala) have only one issued EPC. In the case of multiple EPCs on the same property, most are observed to be similar buildings with duplicate or highly similar EPCs, representing row/terraced houses or arrays of similar MFH. The cadastral data is assigned to the EPC with the corresponding property address. This ap-

proach introduces some uncertainties, but is assumed to still create a useful estimate of building parameter values across the building stock.

The socio-economic and statistical data is available in coordinate squares, 250×250 m in size, aggregated so that the values for the square are the total and mean of all residents with a registered address that are within the boundaries of the square. Given the coordinates of the square, it was possible to connect to the other data sets through the GIS coordinates. However, since the data are aggregated it had to be distributed on the buildings in the evaluated data set, which then could be done through spatial interpolation. From these values, a coordinate point grid was created, and the values belonging to each square were assigned to the centre point of the square. From this grid, values were interpolated to form a finer grid that was used to assign values more accurately for any point that is located between the original square grid points, i.e., all buildings with assigned coordinates.

Before any interpolation process, outliers need to be removed. If some values are disproportionately large, this will affect the neighbouring values significantly. For a normal distribution, the well-known Empirical Rule or 3σ-rule can be used to determine a reasonable confidence interval or to identify outliers [51]. However, the evaluated parameters are not normally distributed and contain some exceptionally remote outliers (up to 25σ), so for this study outliers are removed if they are outside 5 standard deviations (or 5σ) from the mean. According to Chebyshev's Theorem, at least 96% of values will be found within 5σ for any distribution [52].

The grid size for the spatial interpolation was 50 meters. The grid size is a variable that will be a compromise between accuracy and computational time; a smaller grid size would give more accuracy when assigning values to the buildings, but at a higher computational cost.

The socio-economic parameter values were later assigned to all buildings in the data set. The coordinates of each building were assumed to be its centre (calculated centroid of the building's geometric shape). Each building was then assigned the parameter values in the interpolated layers that are nearest (Euclidian distance) to the building centre coordinates.

2.4. Data cleaning

2.4.1. Outliers

Outliers were treated in a similar way as for the statistical values (see Section 2.3), where entries with parameter values ±5σ from the mean were removed for all clustering parameters except age group and share of heating systems.

To remove outliers for the construction year may seem uncalled for, but the process eliminates faulty entries if present and is useful for the Gotland building stock, as some buildings are exceptionally old (built in the 12th century), which may unnecessarily skew the clusters and means.

2.4.2. Normalisation

The *k*-means algorithm is highly dependent on Euclidian distance, so the outcome of the method is sensitive to the magnitude of the clustering parameters, as larger numerical values will overshadow characteristics with smaller numerical values [11]. The order of magnitude will be different for all clustering parameters, thus they need to be standardised or normalised to be comparable in the same range. This will prevent any unintended skewing of the results and guarantees that no specific feature is given any particular weight to the disadvantage of another [14].

Feature scaling, or min-max normalisation, is a method used when non-normally distributed data needs to be scaled to a fixed range, or to allow operations to be performed on all points at the same scale [14]. It is defined as:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{1}$$

and is often preferable to standardisation, such as Z score methods, for *k*-means clustering parameters [53].

2.5. Clustering algorithm

Based on the literature review (Section 1.2), *k*-means is selected as the clustering technique for the building stock segmentation process in this paper.

The *k*-means clustering method is a widely used technique, where the goal is to choose k cluster centres in order to minimise ϕ, the total squared distance between each point and its closest centre [54]. Ideally, the algorithm should result in clusters whose members are close to each other (or have a high degree of similarity) and are well separated. The technique divides n observations into k non-overlapping clusters, where k centres c are chosen to minimise the total squared distance, represented by the criterion function ϕ:

$$\phi = \min_{c \in C} \sum_{i=1}^{k} \sum_{x \in X} ||x - c_i||^2, \tag{2}$$

where x represents a data point within the set $X \subset \mathbb{R}$, c_i is the centre of cluster i, and C is the set of cluster centres [13,55].

The *k*-means algorithm works as follows:

1. Use the *k*-means++ algorithm for cluster centre initialisation (described in detail by David and Vassilvitskii [54]).
2. Compute the point-to-cluster-centroid distance of all observations with the squared Euclidean distance metric.
3. Assign each observation to the cluster with the closest centroid, or individually reassign observations to a different centroid if the reassignment decreases the total sum of distances.
4. Compute the average of the observations in each cluster to obtain k new centroid locations.
5. Iterate steps 2-4 until cluster assignments do not change (or a maximum number of iterations is reached).

The *k*-means++ algorithm selects an observation uniformly at random from data set X, which means that the cluster centres are initialised differently which each run. Ideally, the finalised clusters will be identical regardless of centroid seed and with identical total sum of distances. Usually, however, with less defined clusters or if the clustering algorithm does not converge, the finalised clusters will differ depending on centroid initialisation. To minimise that risk, the algorithm has an assigned number of replicates $r > 0$ where r different sets of seeds are selected. The clustering algorithm then proceeds with the replicate that resulted in the lowest local minimum, i.e. the lowest total sum of distances.

The algorithm was implemented in the MATLAB environment, building upon the available *kmeans* function [56].

2.6. Determination of the optimal number of clusters

Determining the optimal number of clusters, the value of k, is one of the most common issues using the *k*-means method. This is agreed upon in the reviewed literature, and a comparison with other UBEM approaches not using cluster analysis in current research gives no further insights, as the number of archetypes can vary greatly, from 3 [57] to 99 [58]. The basic idea when solving the issue is to run the algorithm for a range of k values, and evaluate a cluster quality index for each run. A larger number of clusters (and associated representative buildings) may guarantee a higher accuracy and representation of the building stock, but with increased computational time, effort, and resources required for model creation [14].

2.6.1. Elbow method

One of the first methods, and most used, is the elbow method [59]. The method involves plotting an error metric against the number of

clusters, and utilising the resulting graph to determine the optimal value of clusters. The goal is to have the error value as low as possible (for as few clusters as possible), as it indicates more homogeneous clusters. However, when the number of clusters k increases, the error value generally decreases. When there are as many clusters as there are data points in the analysed dataset, it will be equal to zero. An inflection point ("elbow" or "knee") in the graph thus indicates the number of clusters where increasing the value of k will not improve the results significantly, implying that the value of k at the inflection point is the optimal number of clusters.

Criticism against the method has stated that it is too simple, and that there is a significant risk of not being able to determine the optimal value of k because the resulting error curve is not always clear [59,60]. The elbow method has since largely been substituted for different approaches, including silhouette analysis [61], optimising certain information criteria [62], and the gap statistic [63]. Some studies have used the R package NbClust [64] which proposes the best clustering scheme by providing 30 indices for determining and evaluating the number of clusters and by varying all combinations of number of clusters, distance metrics, and clustering methods (including *k*-means). The method, however, has a high computational cost, and for Borges et al. [17] it resulted in only 2 clusters for each segmentation, and for Ghiassi and Mahdavi [11] the majority of scenarios had 6 clusters, a lowest-limit number deterministically defined beforehand in order not to have too few clusters; this indicates that the NbClust method tends towards low number results.

For the elbow method, a variety of error metrics can be used to find the elbow to ascertain the value of k. Therefore, if the method can be extended with new approaches, it could therefore still prove a valuable tool for cluster analysis.

2.6.2. Error metric for the elbow method

Halkidi et al. [13] assess a wide variety of clustering validation techniques and trends. One of the more promising evaluated validity indices is the root-mean-square standard deviation (RMSSTD). Using the elbow method and the RMSSTD metric has been successfully used to determine the number of clusters for archetype building development [19], to cluster buildings for the optimisation design of integrated energy systems [65] and for building energy benchmarking [66]. The metric is defined as

$$RMSSTD = \left[\frac{\sum_{i=1}^{k} \sum_{j=1}^{d} \sum_{q=1}^{n_{ij}} (x_q - \bar{x_j})^2}{\sum_{i=1}^{k} \sum_{j=1}^{d} (n_{ij} - 1)} \right]^{\frac{1}{2}}, \tag{3}$$

where k is the total number of clusters, d is the total number of parameters, x is a data point, $\bar{x_j}$ is the mean of data values of j dimension, and n_{ij} is the number of data values of j dimension that belong to cluster i. For all further calculations or descriptions in this paper, the i dimension indexes clusters, and the j dimension indexes parameters.

In this case study the number of data points in each cluster is equal for all parameters, $\sum_j^d n_{ij} = d n_{ij}$, which gives

$$\sum_{i=1}^{k} \sum_{j=1}^{d} (n_{ij} - 1) = d(\sum_{i=1}^{k} n_{ij} - 1), \tag{4}$$

and thus

$$RMSSTD = \sqrt{\frac{\sum_{i=1}^{k} \sum_{j=1}^{d} \sum_{k=1}^{n_{ij}} (x_k - \bar{x_j})^2}{d(\sum_i^k n_{ij} - 1)}}. \tag{5}$$

2.7. Evaluation of clustering results

2.7.1. Identification of cluster centroids and building archetypes

When applying cluster analysis, the archetype definition can differ. When using *average* buildings, the archetype is a virtual building with all characteristics equal to the cluster centroid, which is defined as the hypothetical mean value for each parameter in each cluster. When using *sample* buildings, the archetype is the existing building that is closest for all parameters to the average building [18]. In this paper, the term archetype refers to sample buildings. To find archetype buildings for each cluster, the sample building can be defined by finding the EPC building with the shortest total distance for all parameters to the cluster centroid. This is found by calculating the Euclidean distance between the centroid and all the EPCs in the cluster, using a nearest point-search algorithm [67].

The *representativeness* index can be used to evaluate the level of similarity between the virtual and the actual archetype building [19]. If the closest EPC has the exact same characteristics for each parameter as the cluster centroid, the representativeness would be 100%. The metric is defined so that only fairly high values (>95%) indicate a representative cluster; for Gangolells et al. [19] the values ranged from 76.% to 100%. As the mean values (and spread) of the building parameters are useful for visualisation and could be important for the next steps in UBEM, the representativeness could be a useful way of determining the stability of the clustering results. The representativeness is also different for each cluster, which makes the metric a useful tool for comparing resulting clusters with each other.

The representativeness for each parameter can be expressed as

$$R_j = \frac{\sum_{j=1}^{d} w_j (1 - |p_{j,cent} - p_{j,arch}|)}{\sum_{j=1}^{d} w_j}, \tag{6}$$

where $p_{j,cent}$ is the parameter value for the cluster centroid, $p_{j,arch}$ is the parameter value for the actual archetype building, and w denotes the weighting coefficient [19]. Without weighting coefficient, we have a mean representativeness for each cluster that corresponds to

$$R_i = \frac{1}{d} \sum_{j=1}^{d} (1 - |p_{j,cent} - p_{j,arch}|), \tag{7}$$

which then can be multiplied by 100 to be presented as a percentage.

2.7.2. Cluster spread

The standard deviation was calculated for each parameter in each cluster, where $\sigma_{p,j}$ is the standard deviation for a variable vector p with N scalar observations for each parameter j, defined as

$$\sigma_{p,j} = \sqrt{\frac{1}{N-1} \sum_{n=1}^{N} |p_n - \mu|^2}, \tag{8}$$

where μ is the mean of p:

$$\mu = \frac{1}{N} \sum_{n=1}^{N} p_n. \tag{9}$$

The magnitude of the standard deviation depends on the magnitude of the parameter values, so the standard deviation is not comparable between different parameters (see Section 2.4). For the parameter spread examined in this paper we are also interested in absolute spread, not the difference from the mean, as we want to compare the magnitude of spread for each parameter and cluster. To be able to make a fair comparison between all clustering parameters, the standard deviation is taken after normalisation of the parameters (with feature scaling, see equation 1), so that it becomes an absolute value between 0 and 1. The mean value for all parameters in each cluster can then represent the mean spread for each cluster, as

$$S_{tot,i} = \frac{1}{d} \sum_{j=1}^{d} \sigma_{p,j}. \tag{10}$$

2.7.3. Cluster distinctness

A combination of the two previously discussed metrics may be used to more accurately determine the optimal number of clusters, and more importantly, visualise the differences between each k. For an ideal result, the representativeness should be as high as possible and the total spread as low as possible. Since the goal with clustering is to divide the entire set of data points (i.e., all buildings in the building stock), it is not enough to have several clusters with desirable values, if one or more clusters perform worse. As such, the representativeness and spread should be acceptable over the full set of clusters. A combination metric can be defined to take both these measures into account, for an easier evaluation and visualisation of the clustering results. This novel metric we call the *distinctness* of each cluster.

The distinctness of a cluster is defined to give equal weight to both representativeness and total spread for each cluster. Both the representativeness R and the cluster spread S_{tot} are numbers between 0 and 1, but it is desired for the clusters to have a high representativeness and low spread. We can then define the distinctness D for each cluster as

$$D_i = wR_i + (1 - w)(1 - S_{tot,i}), \tag{11}$$

where w is a weighting factor, which is here set to $w = 0.5$ to give equal weight to both indices. If $w = 1$, the distinctness equals the representativeness, and if $w = 0$ it equals the cluster spread. Regardless, D is also ranging from 0 to 1, where 1 is the highest possible distinctness, i.e., a perfect cluster of 100% representativeness and no cluster spread. To analyse the distinctness for each number of k, so as to determine the optimal number of k, the distinctness has to be determined for all clusters simultaneously in each subsequent run. For this purpose the median value of the distinctness is used, as the distribution is likely to be skewed or have outliers present.

3. Results and analysis

This section presents the results of the clustering procedure and archetype development, and analyses the findings. The discovery of iteration instability in the clustering algorithm is described in Section 3.1. The determined optimal number of clusters and the results of using the new evaluation metrics are presented and analysed in Section 3.2. Section 3.3 later presents the identified archetypes and relates them to the real building stock.

3.1. Clustering iteration instability

The k-means clustering process uses replicates to find and proceed with the replicate resulting in the lowest total sum of distances (see Section 2.5). Even with a high number of replicates, differences in the results could be noted for re-runs even when the process was expected to yield a similar result.

The cluster analysis was run a number of times for an equal set-up of clustering parameters. The resulting error metric values were then plotted together in a graph, where a clear pattern could be noticed, which is seen for 10 subsequent runs in Figs. 4–7. Both metrics display an equal result for each number of clusters, up to a certain point, where the results start to diverge. Increasing the number of k also seems to amplify the difference between each run. A difference in these metrics is an indicator of a difference in the finalised clusters. A minor difference in the metric value is the effect of that a few buildings change which cluster they belong to between runs, and a large difference means that the finalised clusters are exceedingly dissimilar between runs. These results can be interpreted to show that increasing the number of clusters is no longer useful after the divergence point, after which the clustering process becomes unstable. The algorithm will always assign an observation to the nearest centroid, meaning that the data point (building) would ultimately be assigned to the nearest cluster which leads to the most improvement in reducing the criterion function ϕ, even if the improvement is very slight. This could lead to incorrectly defined borders, so in practice buildings are assigned to certain clusters simply because they have to be assigned somewhere, and not because it improves the quality indices. This "forced" assigning could be somewhat arbitrary, and is likely dependent on the initial conditions, since the first step in the algorithm is based on a randomised seeding technique (see Section 2.5).

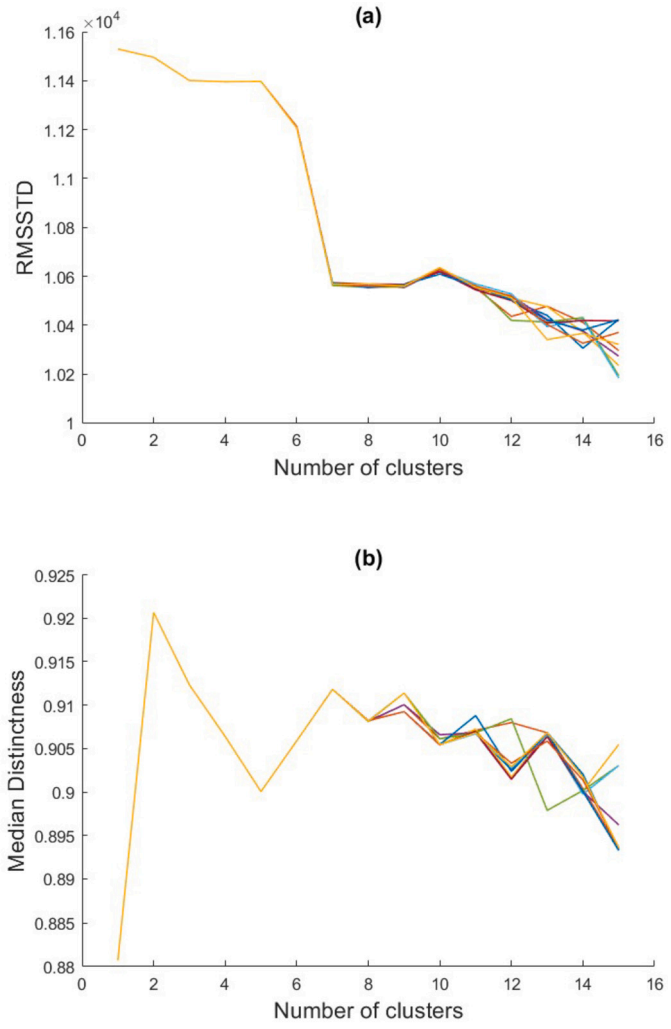


Fig. 4. RMSSTD (a) and median distinctness (b) for SFH in Uppsala.

For each subsequent cluster analysis, the number of replicates per run was set high enough ($r = 100$) to ascertain that the differences in the results are due to this inherent uncertainty of the process, and not that the algorithm itself did not converge.

3.2. Optimal number of clusters

To determine the optimal number of clusters using the elbow method and RMSSTD, the goal is to find the inflection point in the graph. (see Section 2.6) For this analysis, k-means clustering was run on each dataset for $k = 1$ to $k = 15$ and the corresponding RMSSTD value was computed and plotted.

Fig. 4 shows the RMSSTD and median distinctness values for each value of k, for SFH in Uppsala. Each line/colour represent one run of the algorithm, and the graph shows 10 subsequent runs. In Fig. 4a it can be seen that the inflection point is clearly at $k = 7$ for the RMSSTD, which also seems to be the point where the values between each iteration start to diverge. The distinctness (Fig. 4b) shows a spike at $k = 2$, and one again at $k = 7$ before the iterations diverge also for this metric. The second spike at seven clusters also correlates with the clear inflection point for the RMSSTD value. With $k = 2$, only two archetypes

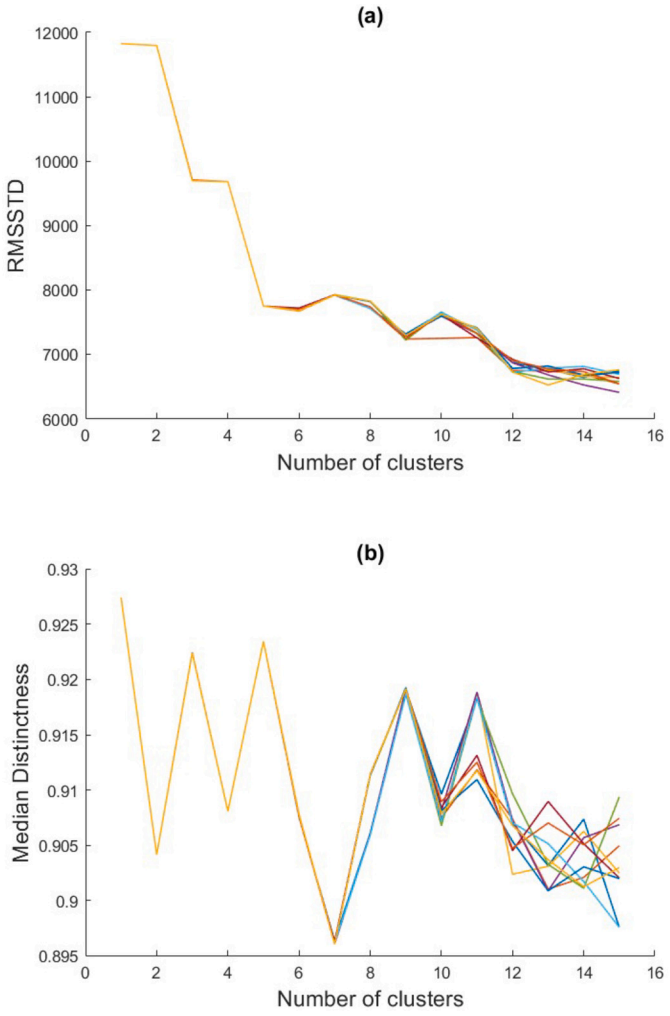


Fig. 5. RMSSTD (a) and median distinctness (b) for MFH in Uppsala.

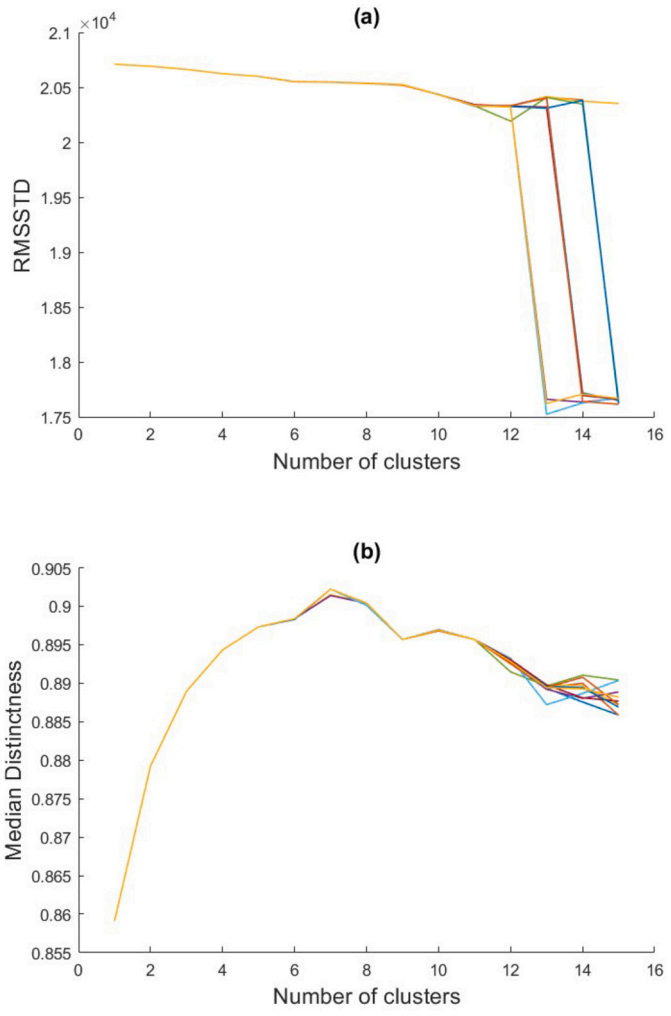


Fig. 6. RMSSTD (a) and median distinctness (b) for SFH on Gotland.

would represent the entire SFH building stock in Uppsala, which could be considered to be too few to be accurate enough, further indicating that $k = 7$ is the optimal value.

Fig. 5 similarly shows the RMSSTD and median distinctness values for MFH in Uppsala. Here we have several inflection points for the RMSSTD, at $k = 3$, $k = 5$ and $k = 9$. The distinctness is also more varying and showing several spikes, interestingly also at $k = 3$, $k = 5$ and $k = 9$, before it diverges. Since the RMSSTD does not decrease much from $k = 5$ to $k = 9$, and the difference between each run increases, $k = 5$ can be argued to be the optimal number.

Fig. 6 shows the RMSSTD and median distinctness for SFH in the Gotland building stock, and similarly for MFH in Fig. 7. Here the RMSSTD shows nearly no change over values of k and has no inflection point, until it diverges significantly. The distinctness however has a clear spike at $k = 7$. For MFH we can instead see a sharp inflection point at $k = 8$ for the RMSSTD but here it is a dissimilar result for the distinctness, as the spike is at $k = 7$. However, it is less defined and the iterations start to diverge earlier, at $k = 6$, and does so significantly for higher values of k. Six clusters are then argued to be optimal for MFH as the distinctness is high and the drop in RMSSTD is not great enough to justify the uncertainty the divergence induces. Increasing to $k = 7$ splits the largest cluster (archetype 6, see Table 4) into two with only slightly different parameter values, but otherwise leaves the other five practically identical. This also supports the argument that six clusters are sufficient for this subset. The results for Gotland indicate that using these two metrics simultaneously can significantly simplify the process of choosing the number of clusters, as one metric may demonstrate a clear result while the other does not.

3.3. Identified clusters and building archetypes

Table 4 shows the identified clusters for SFH and MFH in both building stocks, with their corresponding clustering parameters. The clusters are not ordered in any particular fashion. Uppsala then has 12 archetypes in total, 7 for SFH and 5 for MFH. Both types have one of their largest clusters as the one with the highest distinctness and the smallest cluster with the lowest distinctness. The distinctness does not naturally correlate with cluster size, but if a city has a large number of very similar buildings, they would likely be assorted into the same cluster, which would then have a lower spread. The smaller clusters could have a lower distinctness due to abundant buildings, or outlier buildings that, due to the cluster size, significantly alter the indices.

Gotland has a smaller building stock but has a similar amount of clusters (see Table 4) which could indicate a more diverse building stock, which is true in a sense, due to the amount of historic buildings and that the Gotland building stock is both urban and rural (see Section 2.1). The total spread is significantly higher for all SFH clusters compared to Uppsala, which is due to higher diversity in the clustering parameters but it could also imply that the clustering algorithm is not able to find as homogeneous clusters. The patterns regarding cluster size and distinctness are not as apparent on Gotland even though the same conclusions can be drawn.

Table 4
Final clustering results for the resulting k values, for SFH and MFH in Uppsala and Gotland.

Cluster number	Cluster size [number of buildings]	Representativeness R_i [%]	Cluster spread $S_{tot,i}$ [%]	Distinctness [%]
Uppsala SFH				
1	819	98.3	13.7	92.3
2	1091	98.9	10.6	94.1
3	66	95.9	22.2	86.8
4	546	98.2	15.9	91.2
5	318	97.3	16.4	90.5
6	463	97.8	15.0	91.4
7	215	97.9	16.6	90.2
MFH				
1	1116	98.8	13.1	92.8
2	704	97.6	12.9	92.3
3	130	95.4	20.7	87.3
4	337	96.0	18.1	89.0
5	942	98.5	11.5	93.5
Gotland SFH				
1	747	97.7	16.1	90.8
2	292	96.9	18.8	89.1
3	204	95.9	16.2	89.9
4	390	97.7	16.0	90.9
5	463	97.8	17.5	90.1
6	242	95.8	18.1	88.8
7	412	97.5	16.1	90.7
MFH				
1	581	97.8	13.9	92.0
2	151	98.4	14.7	91.8
3	44	96.7	19.3	88.7
4	76	96.9	19.7	88.6
5	96	96.4	20.5	87.9
6	227	96.3	16.3	90.0

Tables 5 and 6 show the developed archetypes for Uppsala and Gotland with their corresponding cluster parameters and the mean values for each parameter in each cluster. Many archetype values are very close or similar to the mean value of the cluster, especially for the building heating systems. One parameter that could possibly be of extra importance is the energy performance (EP), the normal year corrected energy use per area [kWh/m^2]. Ideally, the EP is the result of many factors that could influence the energy demand, including all or most of the other clustering parameters. A majority of the archetypes have an EP value that deviates 5% or less from the cluster mean, and only one with more than 10%. The construction year of the archetype buildings in Uppsala range from 1948 to 2013. The most newly built SFH archetype also has a very low EP value (52 kWh/m^2), and the earliest built are also the ones with the highest EP values. This correlation between construction year and EP corresponds well with literature [58] and is one reason for why building stocks are almost always segmented by construction year [6,7,9]. However, here we can see that the clustering successfully captures the differences between similarly aged buildings. For example, Uppsala SFH archetypes 2, 3 and 4 are built during the same construction period, but have drastically different other parameter values, especially EP.

For Gotland, the construction year ranges from 1929 to 2007, but with a notable lower likeness between the archetype value and the cluster mean. This is also true for most other parameters, which is also mirrored in the distinctness, as it is generally slightly lower for Gotland compared to Uppsala. However, as the difference is minor, and as the larger clusters have the highest distinctness, it indicates that the archetypes could still be a good representation of the Gotland building stock as well.

The identified archetypes and their most defining characteristics are summarised in Table 7. The defining characteristic of an archetype refers to which aspects that mainly distinguish it from the others, as a qualitative analysis determined from the results in Tables 5 and 6. In Table 7, building energy performance has been labelled from *Very poor* to *Very good*, based on the overall range of values. Building age is based on the year of construction, where *Old* refers to buildings constructed before 1950, *Mid* refers to the period between 1950-1979, *Recent* from 1980-1999 and *Modern* from 2000 and later. To quantify or statistically determine the main distinguishing factor of each archetype, principal component analysis (PCA) or similar feature extraction techniques could be used, but would require a more extensive analysis than can fit into this paper.

To gain a real-world understanding of the cluster characteristics, buildings in a cluster can be observed either on site or using Google Street View [68], and later analysed in a qualitative manner. For a small proof-of-concept study, 20 buildings were selected at random for a random cluster. The selected random cluster was Uppsala SFH 6, where the archetype building was observed to be a 1.5 story house with a garage and a façade of calcium-silicate bricks and wood in a style typical for 1970s Swedish construction. The sample of cluster buildings were observed to share similar characteristics. They were nearly all 1-1.5 story freestanding houses with garages and similar façades. Four clear outlier buildings were noted that stood out in the observation. However, all of them were larger, mansion-like houses that were recently built.

3.4. Geographic distribution

Buildings in city areas are not completely heterogeneous by default, and the built environment in different areas of a city reflects both time periods and social development. New city areas are often constructed as part of large housing projects, resulting in areas with similar or almost identical buildings. Because of this, a satisfactory clustering result

Table 5
Archetype parameters and cluster means (cluster means in parenthesis), Uppsala and Gotland.

Archetype	Construction year [yr]	Building volume [m^3]	Energy Performance [kWh/m^2]	Income [kSEK/p,yr]	Age group	Population density [$persons/km^2$]
Uppsala SFH						
1	1991 (1989)	702 (589)	99 (96)	136 (126)	5.6 (5.5)	919 (1019)
2	1955 (1961)	713 (746)	126 (133)	144 (144)	5.9 (6.0)	699 (676)
3	1948 (1953)	564 (633)	147 (132)	140 (132)	5.0 (5.3)	78 (348)
4	2006 (2008)	393 (477)	52 (54)	102 (106)	5.0 (4.8)	358 (409)
5	1972 (1963)	624 (630)	121 (117)	133 (143)	5.6 (5.7)	615 (469)
6	1970 (1972)	857 (848)	60 (65)	132 (150)	6.0 (5.8)	635 (490)
7	1968 (1971)	676 (638)	74 (86)	143 (136)	6.0 (5.6)	360 (396)
MFH						
1	1975 (1981)	3808 (4027)	146 (138)	131 (136)	5.8 (5.8)	1111 (1106)
2	1948 (1942)	5116 (7042)	149 (137)	173 (178)	5.0 (5.4)	2425 (2434)
3	1968 (1963)	5487 (5625)	73 (70)	113 (125)	5.6 (5.4)	1661 (1545)
4	2013 (2001)	10824 (11451)	80 (86)	144 (152)	5.6 (5.3)	2913 (2600)
5	1976 (1976)	1735 (5041)	136 (131)	87 (100)	5.0 (5.1)	1886 (1897)
Gotland SFH						
1	1971 (1966)	677 (622)	73 (78)	143 (127)	5.9 (5.6)	60 (87)
2	1950 (1959)	496 (616)	105 (114)	145 (123)	5.7 (5.6)	138 (147)
3	1943 (1965)	665 (864)	110 (133)	119 (145)	6.0 (5.9)	512 (479)
4	1950 (1950)	711 (645)	77 (75)	138 (145)	5.8 (6.0)	475 (509)
5	1930 (1942)	506 (648)	174 (175)	128 (123)	5.7 (5.6)	23 (118)
6	1984 (1958)	602 (603)	85 (99)	164 (121)	5.9 (5.6)	102 (106)
7	1968 (1957)	725 (577)	121 (126)	135 (131)	5.6 (5.7)	111 (169)
MFH						
1	1968 (1976)	2478 (3265)	134 (135)	131 (148)	6.0 (5.8)	670 (621)
2	1972 (1982)	3477 (2934)	128 (126)	107 (110)	3.9 (4.0)	205 (294)
3	1980 (1967)	1288 (1949)	101 (127)	134 (122)	5.3 (5.7)	176 (243)
4	1929 (1947)	1318 (2599)	100 (78)	171 (182)	5.9 (6.1)	242 (285)
5	1931 (1968)	1614 (1686)	134 (125)	108 (123)	6.0 (5.8)	289 (230)
6	1968 (1957)	5785 (3608)	104 (134)	200 (248)	5.8 (6.1)	266 (285)

would give clusters that are well distributed geographically; some clusters are more represented in some areas rather than other, and areas will have one cluster that is dominant.

Figs. 8–12 show the geographical locations of the examined buildings within Uppsala and on Gotland, coloured after the finalised clusters, along with the corresponding building archetypes. Figs. 11 and 13 are magnified to show the city of Visby, Gotland in more detail. The building clusters are clearly distinct by location, and the distribution of clusters varies between different areas. Clusters are not distributed by area in any particular pattern on Gotland, and this heterogeneous distribution of clusters is as expected from a rural building stock. Notable, however, is that some clusters are almost only represented in either the city or the countryside. This result confirms the hypothesis that buildings in the same area are more likely to belong to the same cluster. With a knowledge of the study areas, the geographical distribution of the clusters also shows that the algorithm is capable of identifying characteristic patterns in the building stock. For example, it can be noted that student-housing areas and socio-economically disadvantaged areas in Uppsala are clearly represented as belonging to one cluster, MFH 5, which has the lowest income and low mean age. This feature can be used in a future UBEM to evaluate certain areas, such as socio-economically vulnerable areas, more in detail or compare them to other areas more accurately.

For both case studies it can however be noted that no archetype is located in the city centres of Uppsala and Visby (although clusters are present there). Both city centres are densely built with high human activity, as is typical, and have a high share of non-residential buildings. To be able to accurately capture the characteristics of these areas, it could be of interest to deterministically define one or several archetypes for these areas beforehand. Since a large share of the city centre buildings is historical, especially in the city of Visby, to put a weighting factor on the construction year parameter could possibly steer the algorithm to produce archetypes also for these areas.

4. Discussion

4.1. Results and identified archetypes

The fundamental application for identified building archetypes is the inclusion and aggregation into a fully realised UBEM. Recently, Johari et al. [69] developed an accurate UBEM based on available national open data in Sweden, using GIS and EPC data and a simplified deterministic approach for building archetype classification. Using this framework, an UBEM could be developed using the building archetypes from this study. A goal for future research would be to analyse the potential for accuracy improvement using the detailed archetypes, and the use of random sampling from archetype parameter distributions when aggregating the archetypes to large-scale models. The procedure is capable of identifying patterns in the building stock, including socio-economically diverse areas. Future UBEM work could include investigation of energy use patterns in the building stock and evaluation of the potential of UBEM for specific areas or building types. It can also be useful for the simulation procedure itself as socio-economic differences affect building occupancy, and improving occupancy modelling is crucial for improved accuracy. Residential occupancy has been shown to be highly neighbourhood dependent [70], so a way of determining these differences without the need of in-depth data acquisition could be highly useful.

Multivariate cluster analysis can be a significant aid when identifying archetypes based on multiple parameters and open data sets, and at a low cost. The developed procedure allows to partially overcome the issue of data scarcity in urban data sets, which is often discussed in lit-

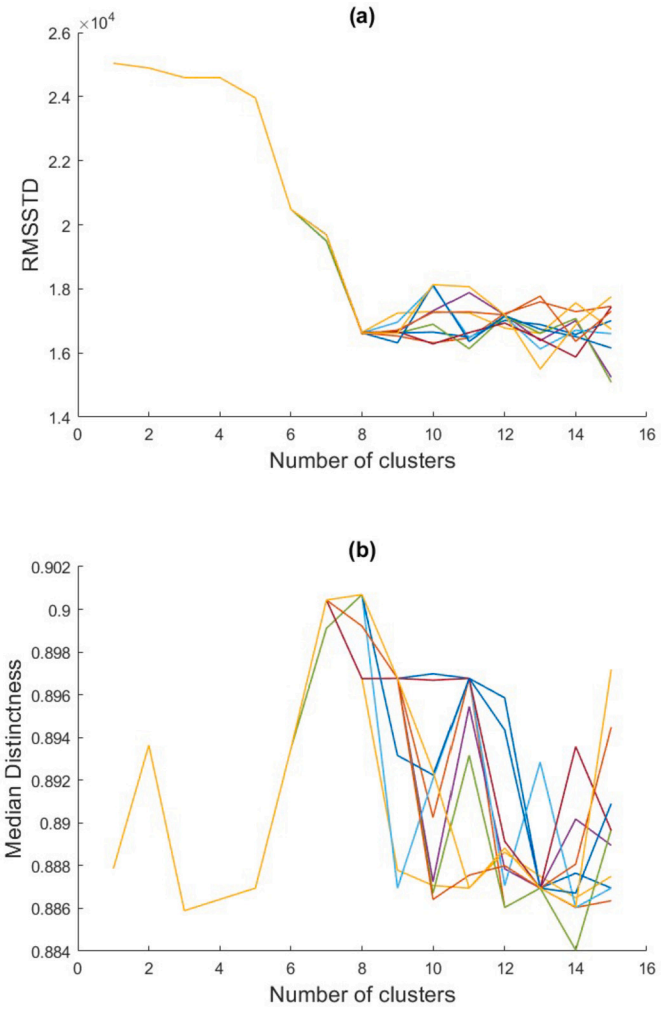


Fig. 7. RMSSTD (a) and median distinctness (b) for MFH on Gotland.

Table 6
Archetype parameters and cluster means (cluster means in parenthesis), Uppsala and Gotland.

Archetype	Share of district heating	Share of boilers	Share of Electric heating	Share of heat pumps
Uppsala SFH				
1	1.0 (0.9)	0.0 (0.0)	0.1 (0.1)	0.0 (0.0)
2	1.0 (1.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)
3	0.0 (0.0)	0.9 (0.8)	0.1 (0.1)	0.0 (0.1)
4	0.0 (0.0)	0.0 (0.0)	0.1 (0.1)	0.9 (0.9)
5	0.0 (0.0)	0.0 (0.0)	1.0 (0.9)	0.0 (0.0)
6	0.0 (0.0)	0.0 (0.0)	0.1 (0.1)	0.9 (0.9)
7	0.0 (0.0)	0.1 (0.1)	0.5 (0.6)	0.4 (0.4)
MFH				
1	0.9 (0.9)	0.0 (0.0)	0.1 (0.1)	0.0 (0.0)
2	0.9 (1.0)	0.0 (0.0)	0.1 (0.1)	0.0 (0.0)
3	0.3 (0.1)	0.0 (0.0)	0.2 (0.3)	0.6 (0.6)
4	0.8 (0.7)	0.0 (0.0)	0.3 (0.2)	0.0 (0.1)
5	0.9 (0.9)	0.0 (0.0)	0.1 (0.1)	0.0 (0.0)
Gotland SFH				
1	0.0 (0.0)	0.0 (0.0)	0.1 (0.1)	0.9 (0.9)
2	0.0 (0.0)	0.2 (0.2)	0.5 (0.6)	0.3 (0.2)
3	1.0 (1.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)
4	0.0 (0.0)	0.0 (0.0)	0.0 (0.1)	1.0 (0.9)
5	0.0 (0.0)	0.9 (0.9)	0.1 (0.1)	0.0 (0.0)
6	0.0 (0.0)	0.4 (0.4)	0.2 (0.1)	0.4 (0.5)
7	0.0 (0.0)	0.0 (0.0)	1.0 (1.0)	0.0 (0.0)
MFH				
1	1.0 (0.9)	0.0 (0.0)	0.0 (0.1)	0.0 (0.0)
2	0.9 (0.9)	0.0 (0.0)	0.1 (0.1)	0.0 (0.0)
3	0.0 (0.0)	0.9 (0.9)	0.1 (0.1)	0.0 (0.0)
4	0.0 (0.0)	0.0 (0.0)	0.1 (0.1)	0.9 (0.9)
5	0.0 (0.0)	0.0 (0.0)	1.0 (1.0)	0.0 (0.1)
6	0.9 (0.9)	0.0 (0.0)	0.1 (0.1)	0.0 (0.0)

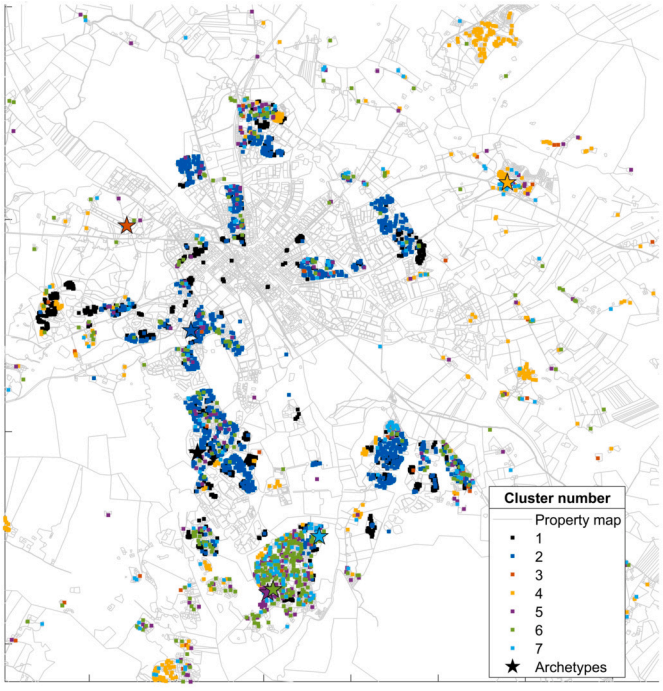


Fig. 8. Identified clusters and archetypes for SFH buildings in Uppsala.

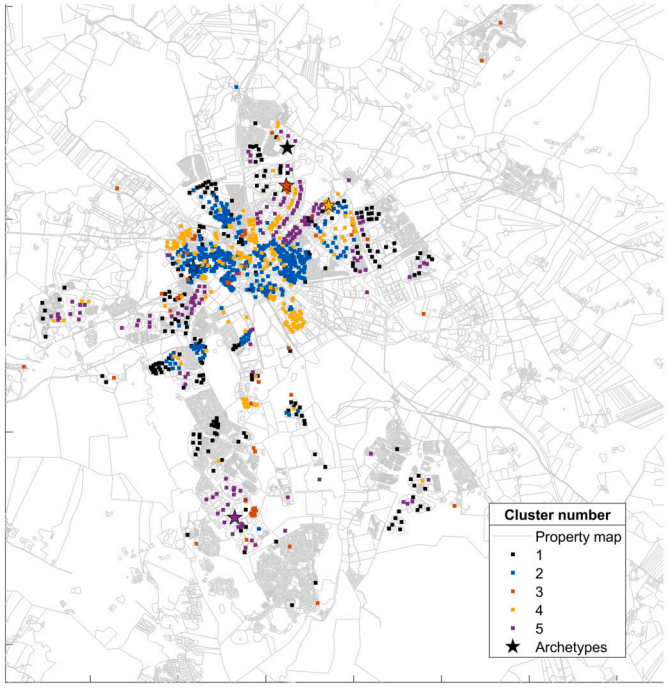


Fig. 9. Identified clusters and archetypes for MFH buildings in Uppsala.

Table 7
Overview characterisation of the identified building archetypes, with energy performance graded from *Very poor* to *Very good*, and Building age categorised as the construction periods *Old* <1950, *Mid* 1950-1979, *Recent* 1980-1999 and *Modern* 2000-present.

Archetype	Energy performance	Heating system	Building age	Other defining indicators
Uppsala SFH				
1	Medium	District heating	Recent	High population density
2	Poor	District heating	Mid	High income
3	Poor	Boilers	Old	Young residents, low population density
4	Very good	Heat pumps	Modern	Low income, young residents
5	Poor	Electric heating	Mid	-
6	Good	Heat pumps	Mid	Large volume
7	Good	Mixed	Mid	High income
MFH				
1	Poor	District heating	Mid	Old residents
2	Poor	District heating	Old	High income, high population density
3	Good	Mixed, with heat pumps	Mid	-
4	Good	Mixed, no heat pumps	Modern	Large volume, high population density
5	Poor	District heating	Mid	Low income, young residents
Gotland SFH				
1	Good	Heat pumps	Mid	Low population density, rural
2	Medium	Mixed	Mid	High population density, rural
3	Medium	District heating	Old	High population density, urban
4	Good	Heat pumps	Mid	High population density, urban
5	Very poor	Boilers	Old	Low population density, rural
6	Good	Mixed	Recent	High income
7	Poor	Electric heating	Mid	Large volume
MFH				
1	Poor	District heating	Mid	High population density
2	Poor	District heating	Mid	Low income, young residents
3	Medium	Boilers	Recent	Low population density
4	Medium	Heat pumps	Old	High income
5	Poor	Electric heating	Old	Low income
6	Medium	District heating	Mid	High income

erature. As it utilises open data and a thoroughly described process the study can be replicated, which we see as a major benefit of the procedure. Using *k*-means clustering of multiple variables, 12 archetypes were identified for the residential building stocks in Uppsala and 13 on Gotland. The identified archetypes are representative of their respective cluster of building data points, but are also diverse among themselves, covering a wide range of building characteristics.

The clustering algorithm worked well with parameters not often used in the field, namely socio-economic indicators and building heating systems. Adding more parameters in the clustering process decreases the overall accuracy as more uncertainties are introduced, and the representativeness is measured against additional parameters. Running the procedure with a higher number of parameters also results in a higher number of archetypes, compared to running with fewer parameters, but this is expected as the building stock then can be divided based on more factors.

The use of spatial interpolation of statistical data also introduces uncertainties. In addition to uncertainties in the underlying data, uncertainty in spatial models is dependent on the resolution of spatial data, the spatial autocorrelation of the parameter values, that is the degree of correlation or similarity between nearby observations, and the interpolation grid size [71]. The interpolation is based on data points representing the parameter mean values for all buildings within 250×250 m squares, which could eradicate the inherent differences between buildings within this area, and limits the resolution of spatial data to 250 m. The use of spatial interpolation for this procedure relies on the assumption that neighbouring areas are more similar (see Section 3.4) also regarding socio-economic indicators, in this case corresponding to a high degree of parameter autocorrelation. Statistical data that can be tied to individuals, households or buildings would evidently improve the accuracy significantly, but are not open data, which means using them would go against the aim of this study.

The increased number of parameters does give more depth to the identified archetypes as they contain more information about the characteristics of their corresponding clusters, and allows for the division of the building stock based on other aspects than those traditionally used. When analysing the results (Tables 5 to 7) significant patterns can be clearly noted. Building heating system seems to be a decisive factor for both the building EP and the division into archetypes, and the socio-economic parameters are often the divisive factor distinguishing two archetypes with similar heating system and construction period. This could be of importance to future UBEM applications targeting or comparing certain city areas or building types. The importance of the heating system for the EP of a building can also be seen in the results as the EP values are clearly different between the archetypes. This implies that the division into archetypes by building heating systems could be important for the later UBEM stages, as a way to increase model accuracy and to mitigate the performance-simulation gap.

Buildings that are not present in the EPC data set need to be characterised when applying the UBEM over the whole area, where the central step is to assign them into the right archetype. As the socio-economic data is available for the entire area, the differences in these parameters between the archetypes could greatly improve the assigning of these buildings into the most representative archetype. Furthermore, to analyse the main distinguishing factors in depth and their significance for archetype classification, using for example PCA techniques, could be rewarding for future research.

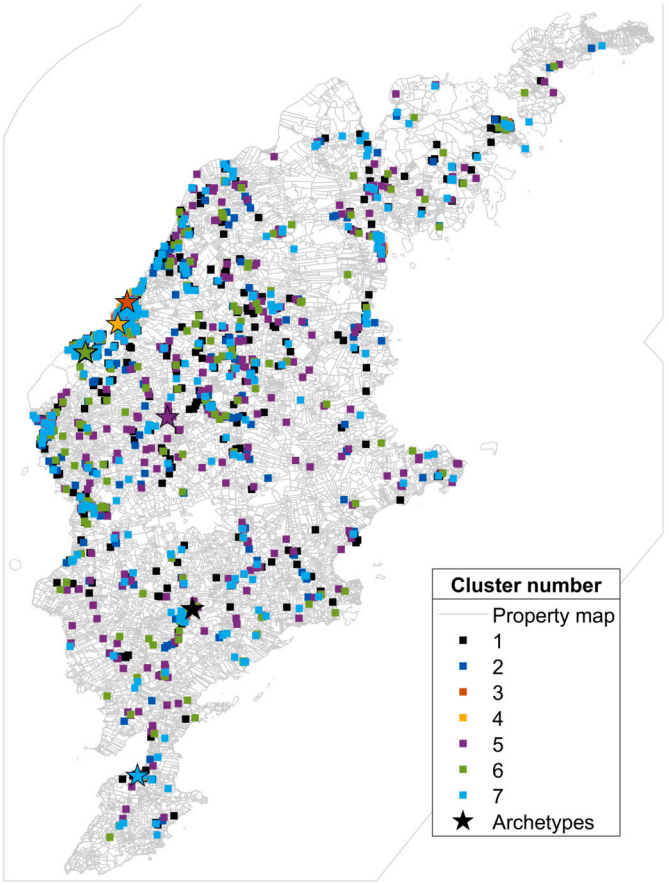


Fig. 10. Identified clusters and archetypes for SFH on Gotland.

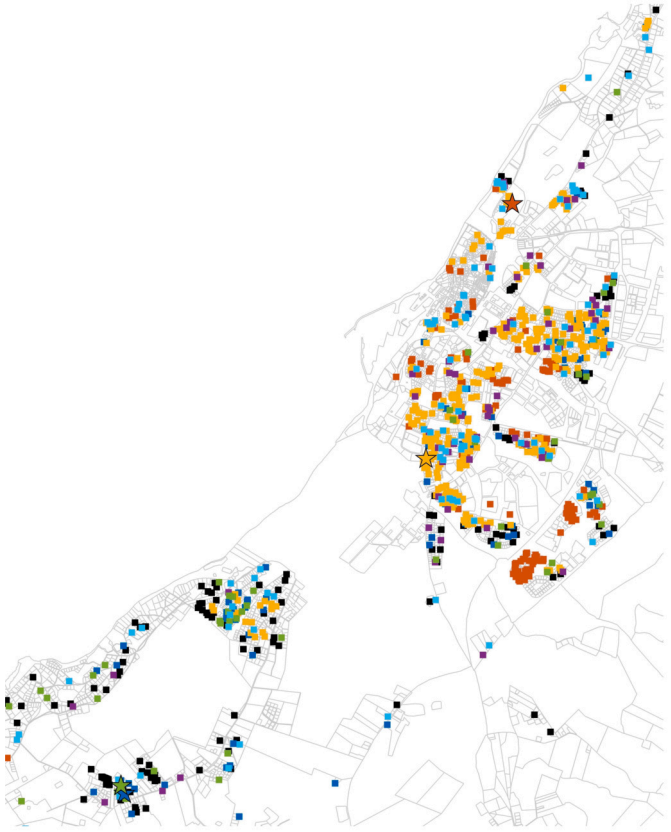


Fig. 11. Identified clusters and archetypes for SFH in Visby.

4.2. *Clustering procedure*

The use of *k*-means clustering and the elbow method to identify building archetypes is common in the field. However, the wide selection of clustering parameters utilised in this study, namely building heating systems, socio-economic factors and measured energy performance, is not. Additionally, the analysis of the divergence point of the error metric to determine the maximum amount of stable clusters has to our knowledge not been previously utilised. Most studies either apply an overly simple analysis and a deterministic way to decide the value of *k*, or apply a computation-heavy, time-consuming analysis of multiple methods and indices. There is also the considerable issue regarding the lack of similar studies that are capable of replication, indicating the importance of studies utilising openly available data sets and an adaptable methodology.

Both the RMSSTD and distinctness indices were shown to be useful tools when determining the optimal number of clusters, complementing each other effectively and providing useful visual representation. However, the approach to determine *k* by identifying the inflection point, or "elbow", in the RMSSTD curve [13,19] does not work if the resulting graph does not have an elbow or if it has multiple elbows. As shown in this paper, the RMSSTD curve could also be misguiding if one does not account for the possible divergence between iterations due to the "forced clustering". This analysis of the divergence between algorithm iterations is important for getting reliable results out of the clustering technique and is possibly completely new to the field. These findings may have potential implications for utilising simple error metrics when evaluating cluster analysis results, which could be further investigated in future research. For validating cluster analysis, it has been pointed out that the degree of consistency in algorithm runs indicates reliability [60] and that for strong confidence in the results, techniques that are

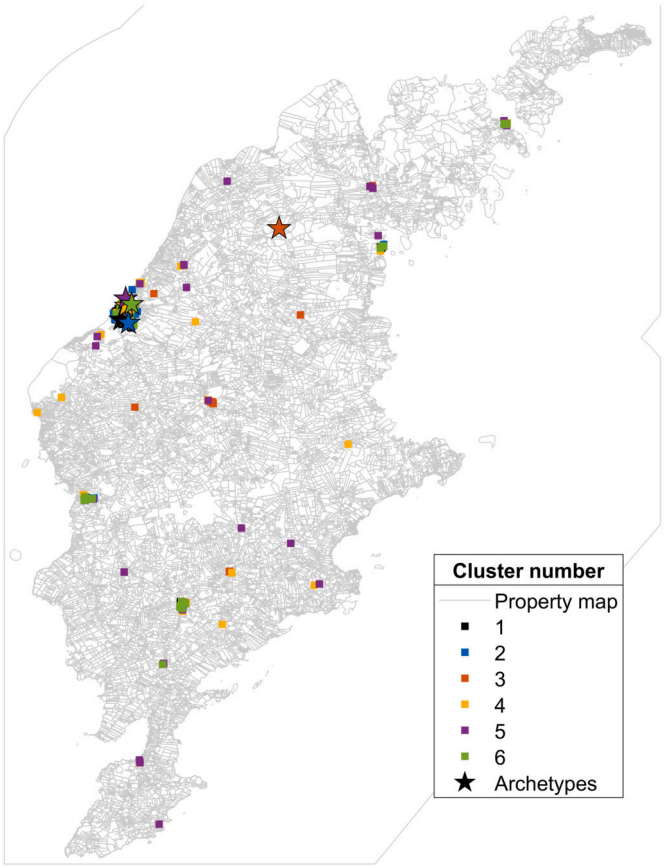


Fig. 12. Identified clusters and archetypes for MFH on Gotland.

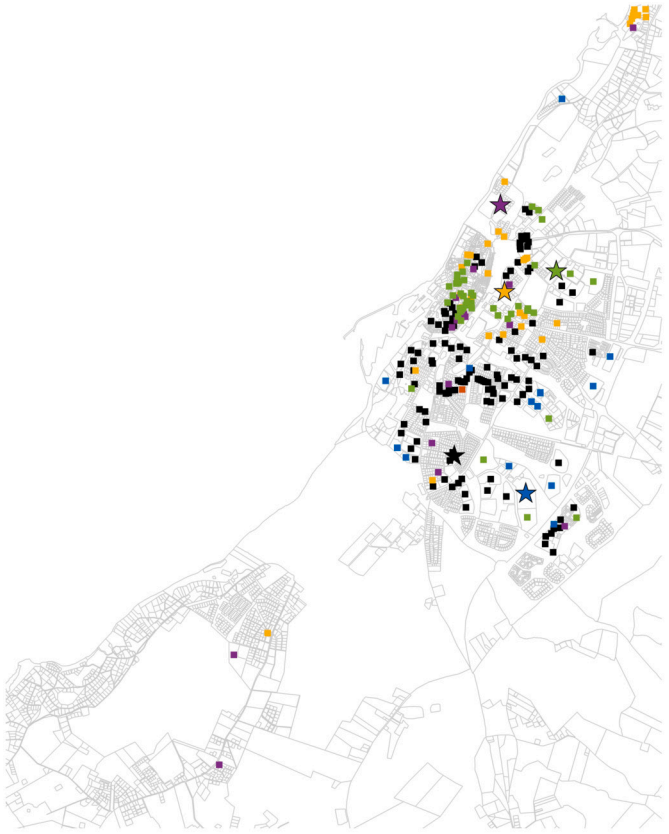


Fig. 13. Identified clusters and archetypes for MFH in Visby.

less subject to the bias of researchers must be used. This reliability is increased by running the algorithm for several iterations and noticing the divergence point, as shown in this paper.

Our proposed index, the cluster distinctness, focuses on the compactness of each cluster (same as RMSSTD) but not on the distance between each cluster. It also focuses more on the mean value for each parameter for each cluster, and puts equal weight on how representative the archetype building is of said cluster. Algorithms that use a large number of different indices to determine the optimal clustering method and number of clusters (like NbClust) exist and work well; however, the computational time for such algorithms is significant and they usually provide no means of visualisation. Every different index also has slightly different targets; depending on what is important for each study, different indices may be the most useful. When used for similar applications, this algorithm does however seem to produce only very low values of k (see Section 2.6), which results in a number of clusters too small for many applications.

4.3. Limitations and areas for further research

The choice of clustering technique and metrics for assessing cluster quality can be subjective; choosing a different method or metric will most certainly yield different results since the criteria for defining the optimal clusters would change. The choice of method and metric in this paper was made based on a literature review of similar studies, where k-means and RMSSTD where shown to be the most promising clustering method and quality metric, respectively. Since the aim of this study was to evaluate the cluster analysis of an extended number of parameters, which was successful using the chosen method, this approach can be taken using alternative clustering methods. A comparison of different clustering techniques and metrics for an extended set of parameters could be useful for future research in the area. A definite evaluation of how different clustering techniques and choices of metrics would impact the performance of UBEMs would require applying the resulting archetypes to large-scale UBEM simulations and comparing the results against measurement data. This is a too extensive analysis to include in this paper, but would be interesting as a future study.

For geometric data, there exist several open standardised geospatial data formats, including CityGML [72] or OpenStreetMap [73], that have recently gained popularity as comprehensive data formats to use for UBEM [10,22]. These formats contain useful information about the urban environment but are not always available for urban areas and usually lack vital building information [22,74]. For the scope of this paper, a 3D representation of the full studied building stock was not needed as the 2D cadastral data provided all vital information for the used approach. For later stages of UBEM development, which includes 3D model development, the archetypes identified in this paper could still be used even when applying CityGML or similar formats for these stages, given that key geospatial characteristics such as building footprint and position are consistent in the respective data formats.

The study was based on EPCs, which provide very useful data but only cover a part of the building stock, creating a sample bias. The share of buildings that do have an EPC is smaller for SFH compared to MFH, even more so for rural SFH, and the study left out non-residential buildings for similar reasons. UBEMs often aim to include all buildings, including non-residential buildings such as offices, schools, and industry. Offices or mixed-use buildings have been included in, or have been the main focus of, similar studies using a clustering approach [14,17,19]. The number of other non-residential buildings (schools, hospitals, etc.) is generally low compared to other use types, so using separate archetypes for these building types is often unnecessary. Future UBEMs can utilise assumptions made from previous studies, or apply individual BEM if metered data is unavailable.

Upscaling of data to city-scale level is necessary for the UBEM process, regardless of using EPCs or other data sources to identify and characterise the archetypes, and is highly important for further research following the developed methodology. For this process there are two major uncertainties; to assign all buildings the appropriate archetype and characterisation, and the accuracy of archetypes in terms of representing the entire building stock. The insufficient coverage of some data is affecting the accuracy of the archetypes, which also is the focus of this paper. For the later-stage UBEM processes, measures can be taken to ensure higher accuracy or data coverage since it is known for which buildings data is missing. Data for buildings without EPCs could be obtained by resorting to other sources, for example by collecting new data through inventory studies or by manually defining archetypes for these buildings in a deterministic fashion. Two separate analyses could also be performed and later combined to identify archetypes; one for EPC parameters and one for building volume and socio-economic parameters, which are available for the whole building stock with the approach used in this paper. To ensure the identification accuracy also for rural SFH, a better insight into the actual composition of the building stock is needed. It is necessary to ascertain if our assumption is correct, or if the distribution of building characteristics is skewed compared to reality. A study performing a more detailed inventory of a selected area, for example a countryside parish, and comparing with data and archetypes from this study, would be helpful in validating the methodology. We have made the assumption that the EPCs are a good representation of the building stock despite the smaller sample size and that the distribution of parameter values is statistically accurate on an aggregate level. Furthermore, the EPCs in the EU are about to be extended and harmonised between all member states, as pointed out in Section 1.3. With better comparability there will be even more areas with the ability to adapt the procedures developed in this study, and the accuracy of the approach will increase with increasing EPC coverage across the building stocks. Additionally, even as the coverage per capita in Sweden is increasing, it is not currently in the top five of Europe [34] which indicates that the accuracy would be even higher for other European building stocks.

The developed methodology represents the first step towards developing a novel UBEM approach for the building stock in Sweden or any country with similar availability of open data sets. Following the scope of this paper, the successive strategy is to apply this method and the resulting archetypes in a large-scale UBEM, where the analysis, description and characterisation of the identified archetypes are left to future studies.

5. Conclusions

This paper presented a replicable methodology to identify archetype buildings by using a simple yet reliable algorithm using *k*-means clustering, which was applied to two case studies of the Swedish residential building stock. The proposed procedure is computationally fast and can be made automated which can significantly simplify database prehandling efforts. Furthermore, it requires only open data; this makes the methodology easy to apply to different areas and thus useful for comparing results, which is proven by performing the analysis on two large data sets. The developed procedure can be applied in future UBEM studies of any area with similar availability of open data sets. Additionally, the relevance of the approach is only expected to rise as it will have an increased application field along with the current increase in availability and coverage of energy performance certificates in the EU. Combined with a reliable approach to aggregate the result for the buildings that are not present in the used data set, these studies can apply the methodology to identify archetype buildings that are representative of the studied building stock, which can later be used to develop a reliable and accurate UBEM.

The algorithm is able to identify visible patterns in the studied building stocks, which are transferred to the building archetypes. With this approach, archetypes display differences in building heating systems and can represent socio-economically differentiated areas, and the archetypes are characterised by parameter distributions which could be useful for aggregation to larger scale. When determining the optimal number of clusters an augmented version of the simple elbow method is sufficient to develop representative building clusters. Complementary to finding the elbow for the error metric, the cluster distinctness and divergence point for determining the maximum number of clusters are evaluated. The resulting clusters are well defined, both parameter-wise and by geographic distribution. This indicates that the methodology better captures the diversity of building characteristics within the building stock, including socio-economic parameters, compared to typical deterministic segmentation.

In this paper, we have shown that this approach is useful for identifying building archetypes, that the method is reliable and easily adapted, and that the resulting archetype buildings are representative for their respective clusters – mainly in terms of the numerical metrics employed but also as suggested by a small qualitative analysis – which indicates an accurate representation of the studied building stock.

CRediT authorship contribution statement

Lukas Dahlström: Conceptualization, Methodology, Writing – original draft, Software. **Fatemeh Johari:** Methodology. **Tor Broström:** Supervision, Writing – review & editing. **Joakim Widén:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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